Name
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Course

A Few More Things

Contents

Using a Calculator to Evaluate Cosecant, Secant, and Cotangent Functions

Find the values of the following expressions with a calculator. Round to the nearest thousandth.

① $\csc 73^\circ = \dfrac{1}{\sin 73^\circ} \approx \boxed{1.046}$

② $\sec 35^\circ = \dfrac{1}{\cos 35^\circ} \approx \boxed{1.221}$

③ $\cot 217^\circ = \dfrac{1}{\tan 217^\circ} \approx \boxed{1.327}$

④ $\csc 125^\circ = \dfrac{1}{\sin 125^\circ} \approx \boxed{1.221}$

⑤ $\sec 149^\circ = \dfrac{1}{\cos 149^\circ} \approx \boxed{-1.167}$

⑥ $\cot 39^\circ = \dfrac{1}{\tan 39^\circ} \approx \boxed{1.235}$

Using a Calculator to Solve Trigonometric Equations Involving the Cosecant, Secant, and Cotangent Functions

7. Solve if $0 < \theta < 360°$

$$\csc\theta = 5$$

$$\frac{1}{\sin\theta} = 5$$

$$\sin\theta \cdot \frac{1}{\sin\theta} = 5 \cdot \sin\theta$$

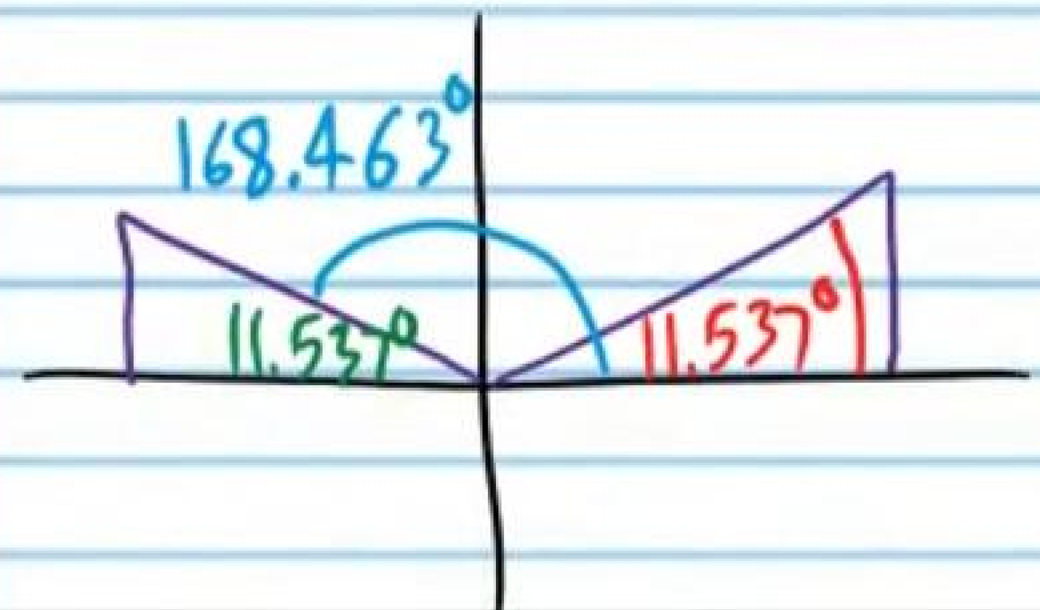
$$1 = 5\sin\theta$$

$$\frac{1}{5} = \frac{5\sin\theta}{5}$$

$$\frac{1}{5} = \sin\theta$$

$$\theta = \sin^{-1}\left(\frac{1}{5}\right)$$

$$\theta \approx 11.537°$$



$$\boxed{\theta \approx 11.537°, 168.463°}$$

8. Solve if $0 \leq W \leq 360°$

$$-3 = 11\cot W + 5$$

$$-8 = 11\cot W$$

$$-\frac{8}{11} = \cot W$$

$$-\frac{8}{11} = \frac{1}{\tan W}$$

$$\tan W\left(-\frac{8}{11}\right) = \frac{1}{\tan W} \cdot \tan W$$

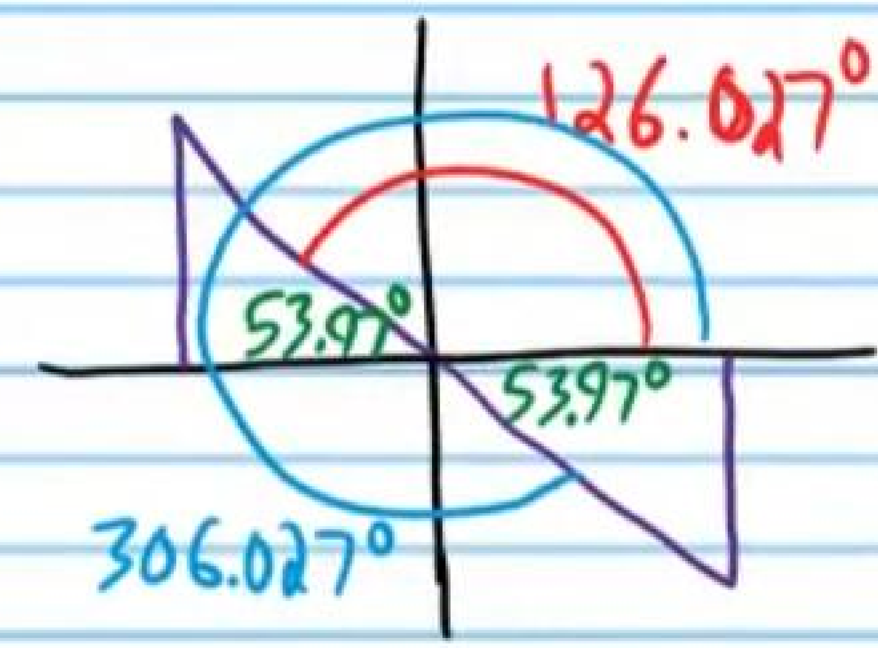
$$-\frac{8}{11}\tan W = 1$$

$$\frac{-\frac{8}{11}\tan W}{-\frac{8}{11}} = \frac{1 \cdot 11}{-\frac{8}{11} \cdot 11}$$

$$\tan W = \frac{11}{-8}$$

$$W = \tan^{-1}\left(-\frac{11}{8}\right)$$

$$W \approx -53.9726°$$



$$\boxed{W \approx 126.027°, 306.027°}$$

⑨ Solve if $0 \le \alpha < 2\pi$.

$$7\sec\alpha - 9 = 10\sec\alpha - 31$$

$$-9 = 3\sec\alpha - 31$$

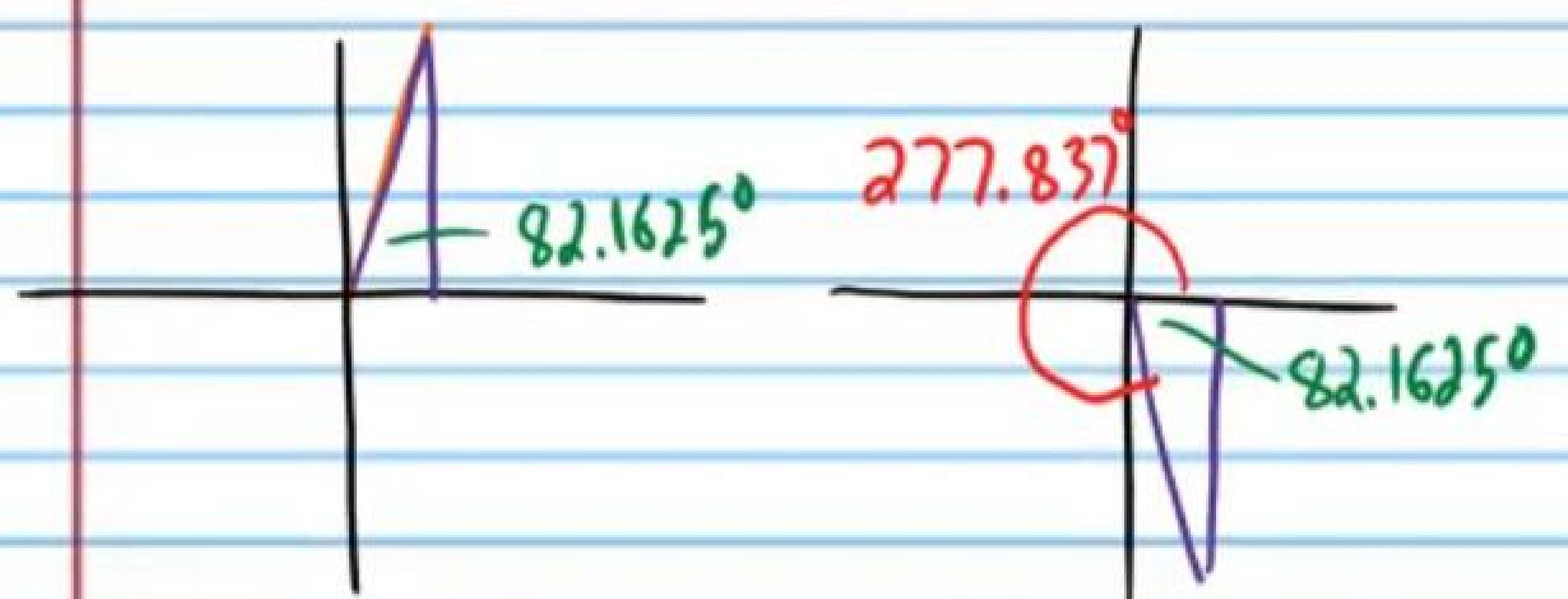
$$22 = 3\sec\alpha$$

$$\frac{22}{3} = \sec\alpha$$

$$\cos\alpha = \frac{3}{22}$$

$$\alpha = \cos^{-1}\left(\frac{3}{22}\right)$$

$$\alpha \approx 82.1625^\circ$$



$$\alpha \approx 82.1625^\circ\ldots \times \frac{\pi}{180^\circ} \approx \boxed{1.434}$$

$$\alpha \approx 277.837^\circ\ldots \times \frac{\pi}{180^\circ} \approx \boxed{4.849}$$

⑩ Solve if $0 \le \beta \le 360^\circ$

$$\cot\beta = 0.64$$

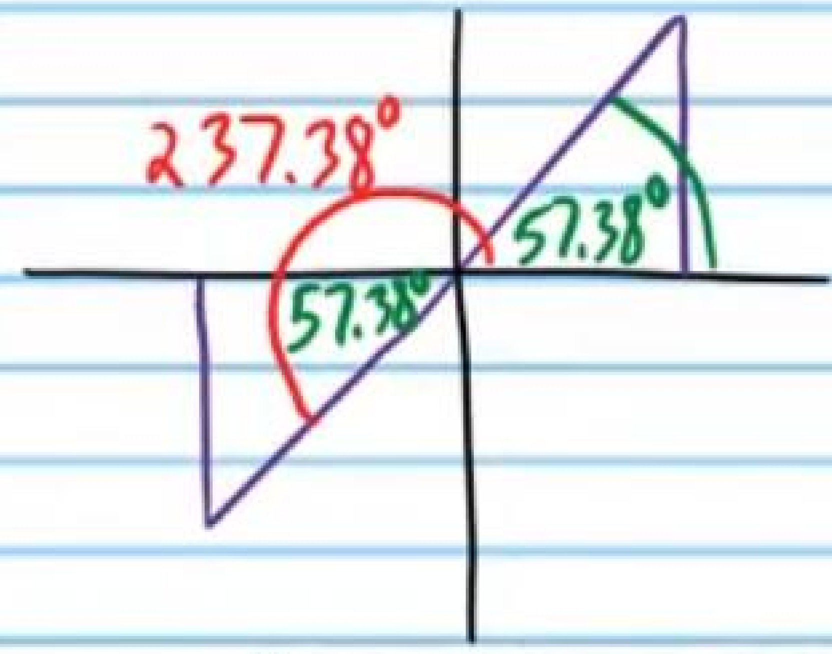
$$\frac{1}{\tan\beta} = 0.64$$

$$\tan\beta = \frac{1}{0.64}$$

$$\tan\beta = 1.5625$$

$$\beta = \tan^{-1} 1.5625$$

$$\beta \approx 57.38^\circ\ldots$$



$$\boxed{\beta \approx 57.38^\circ, 237.38^\circ}$$

☆ Your calculator will not give you the reference angle.

(11) Solve if $0 \le A < 2\pi$.

$$-8\sec A + 3 = 67$$

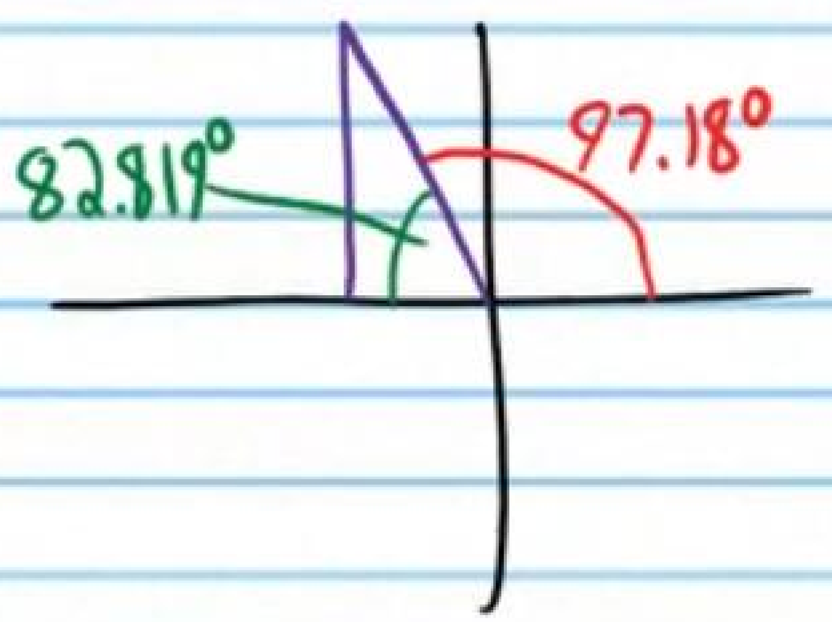
$$-8\sec A = 64$$

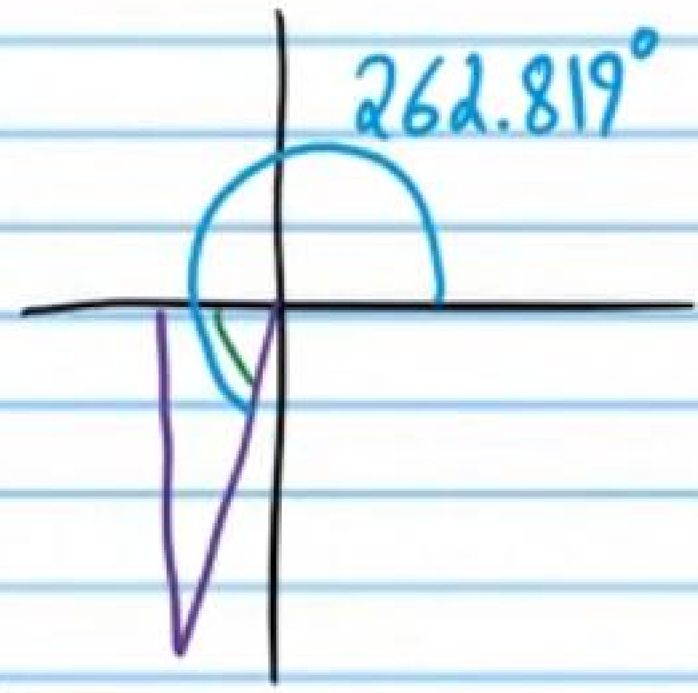
$$\sec A = -8$$

$$\cos A = -\frac{1}{8}$$

$$A = \cos^{-1}\left(-\frac{1}{8}\right)$$

$$A \approx 97.18^\circ$$





$$A \approx 97.18^\circ\ldots \times \frac{\pi}{180^\circ} \approx \boxed{1.696}$$

$$A \approx 262.819^\circ\ldots \times \frac{\pi}{180^\circ} \approx \boxed{4.587}$$

(12) Solve if $0 < B < 360^\circ$.

$$-20 + 7\csc B = -\csc B$$

$$-20 = -8\csc B$$

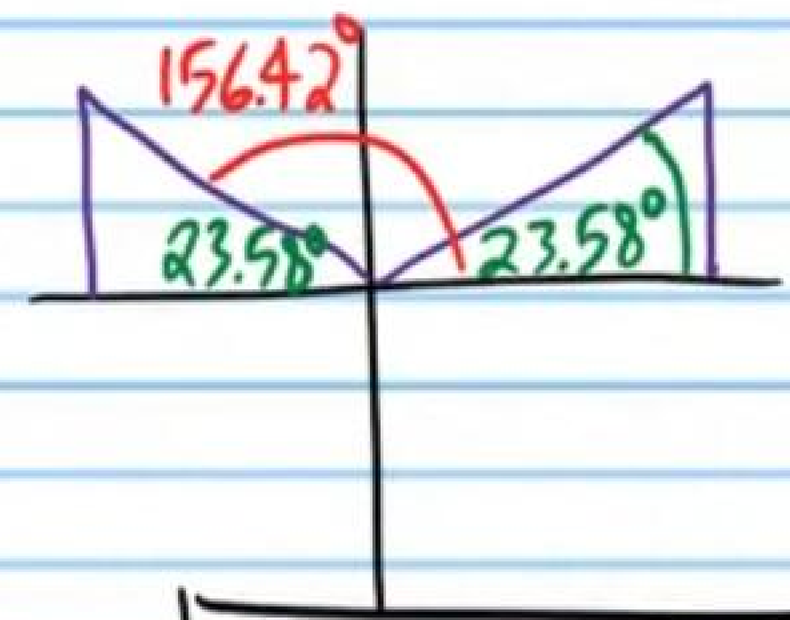
$$\frac{20}{8} = \csc B$$

$$\frac{5}{2} = \csc B$$

$$\sin B = \frac{2}{5}$$

$$B = \sin^{-1}\left(\frac{2}{5}\right)$$

$$B \approx 23.578^\circ\ldots$$

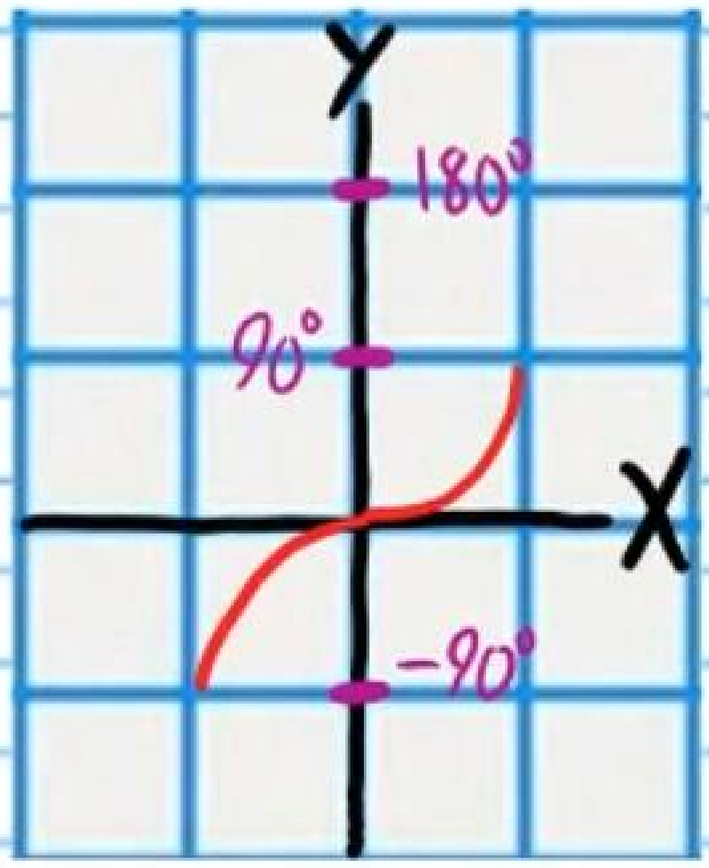


$$\boxed{B \approx 23.578^\circ, 156.4218^\circ}$$

Inverse Trigonometric Functions and Their Graphs

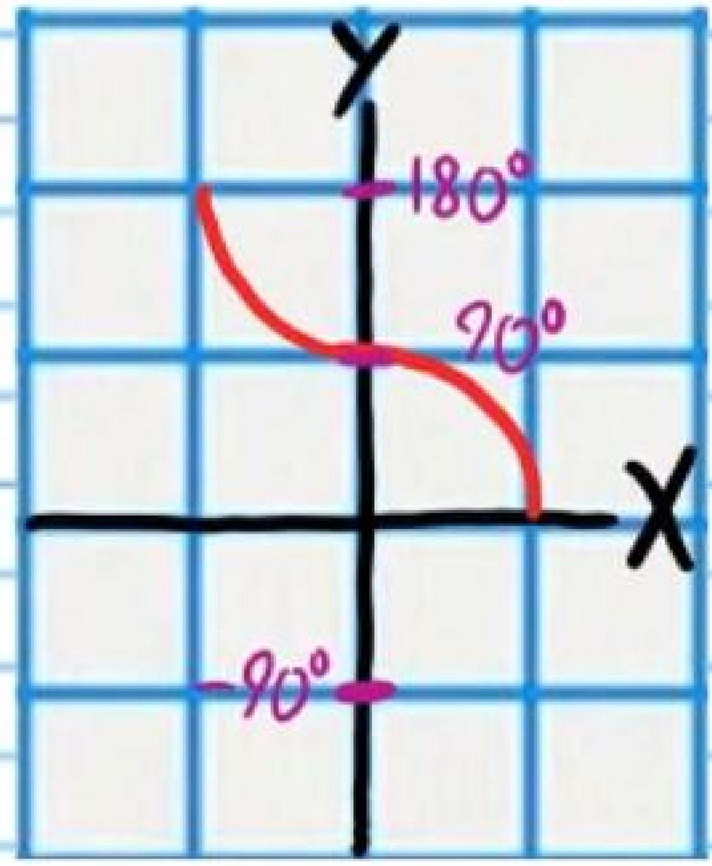
$y=\sin^{-1}x$

Range: $-90^\circ \leq y \leq 90^\circ$



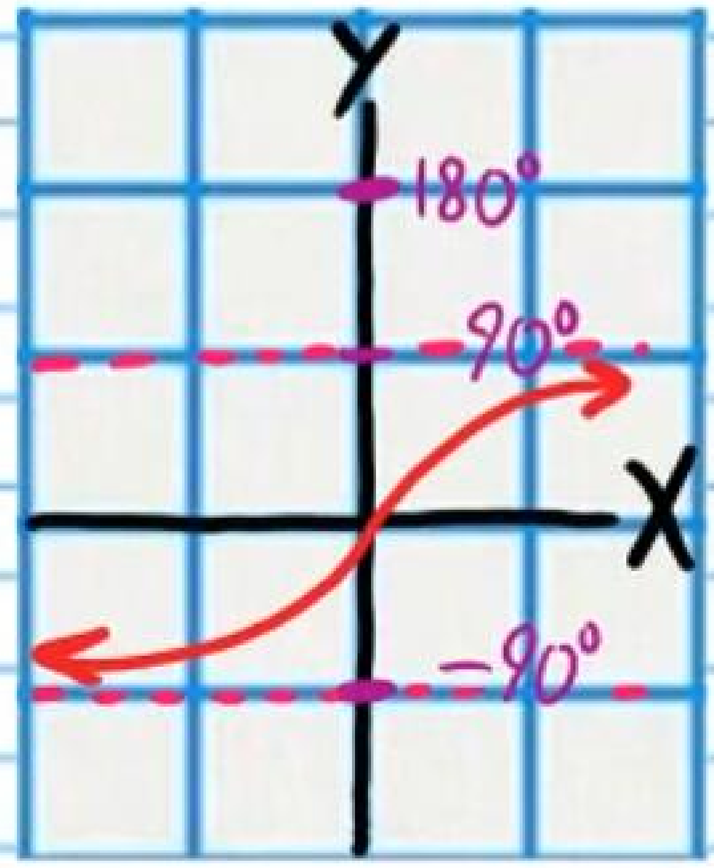
$y=\cos^{-1}x$

$0^\circ \leq y \leq 180^\circ$



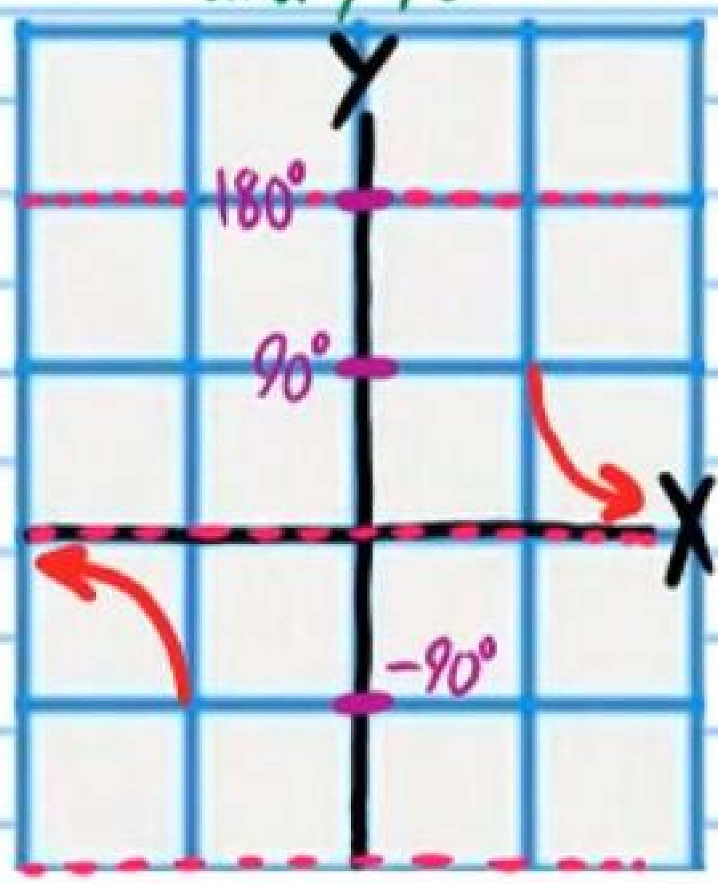
$y=\tan^{-1}x$

$-90^\circ < y < 90^\circ$



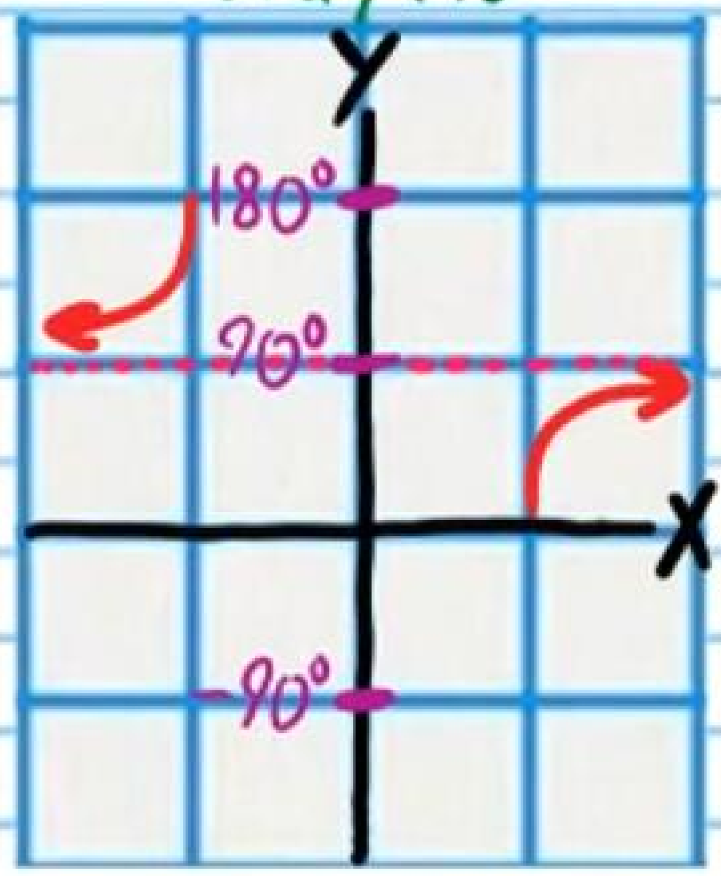
$y=\csc^{-1}x$

Range: $-90^\circ \leq y \leq 90^\circ$ and $y \neq 0$.



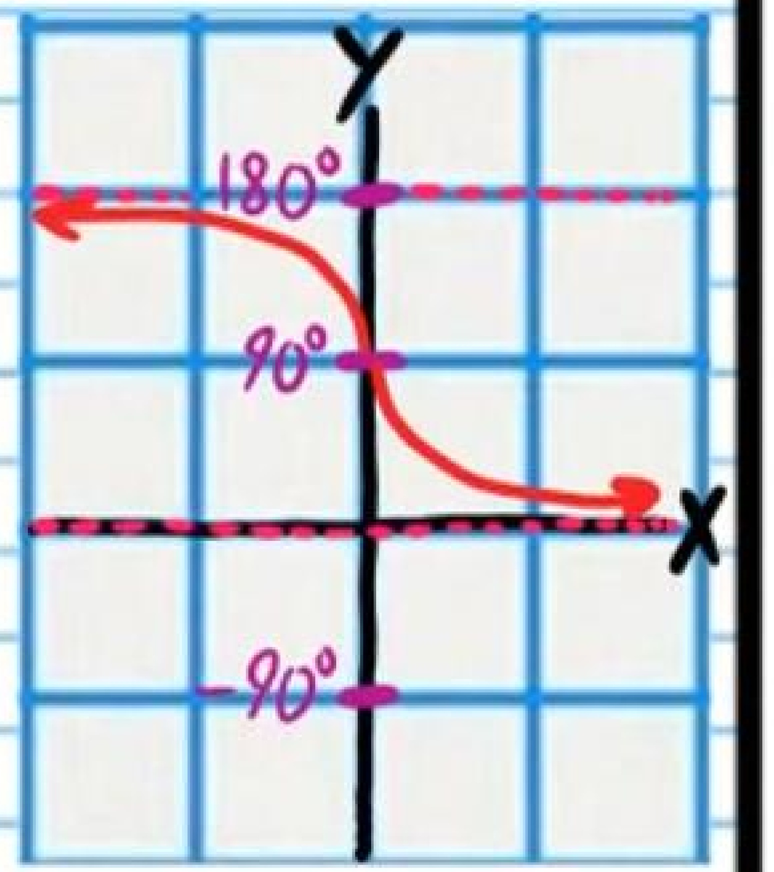
$y=\sec^{-1}x$

$0^\circ \leq y \leq 180^\circ$ and $y \neq 90^\circ$



$y=\cot^{-1}x$

$0^\circ < y < 180^\circ$



Using a Calculator to Evaluate Inverse Cosecant, Secant, and Cotangent Expressions.

Review

Example I) Find the value of $\sin^{-1}\left(\frac{1}{2}\right)$ without a calculator.

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\boxed{\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ}$$

Example II) Find the value of $\sin^{-1}\left(\frac{4}{5}\right)$ with a calculator.

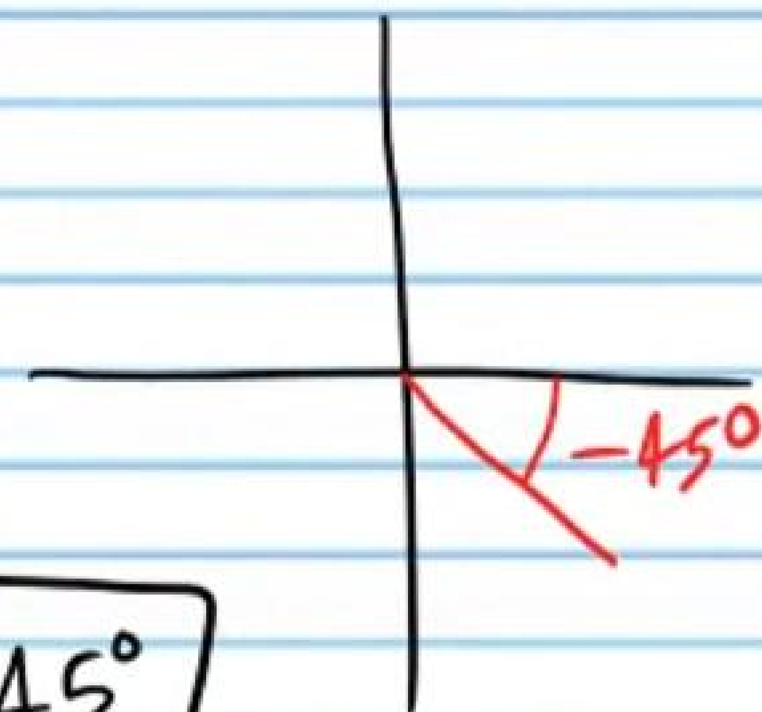
$$\boxed{\sin^{-1}\left(\frac{4}{5}\right) \approx 53.13^\circ}$$

Example III) Find the value of $\csc^{-1}(-\sqrt{2})$ without a calculator.

$$\theta = \csc^{-1}(-\sqrt{2})$$

$$\csc\theta = -\sqrt{2}$$

$$\theta = -45^\circ$$



$$\boxed{\csc^{-1}(-\sqrt{2}) = -45^\circ}$$

Find the values of the following expressions with a calculator. Round to the nearest thousandth.

13. $\csc^{-1}\left(\frac{5}{2}\right)$

$\theta = \csc^{-1}\left(\frac{5}{2}\right)$

$\csc\theta = \frac{5}{2}$

$\sin\theta = \frac{2}{5}$

$\theta = \sin^{-1}\left(\frac{2}{5}\right)$

$\theta \approx 23.578^\circ$

$\boxed{\csc^{-1}\left(\frac{5}{2}\right) \approx 23.578^\circ}$

14. $\sec^{-1}(-10)$

$\theta = \sec^{-1}(-10)$

$\sec\theta = -10$

$\cos\theta = -\frac{1}{10}$

$\theta = \cos^{-1}\left(-\frac{1}{10}\right)$

$\theta \approx 95.739^\circ$

$\boxed{\sec^{-1}(-10) \approx 95.739^\circ}$

15. $\cot^{-1}(3)$

$\theta = \cot^{-1}(3)$

$\cot\theta = 3$

$\tan\theta = \frac{1}{3}$

$\theta = \tan^{-1}\left(\frac{1}{3}\right)$

$\theta \approx 18.435^\circ$

$\boxed{\cot^{-1}(3) \approx 18.435^\circ}$

☆ Your answer should be negative.

16. $\csc^{-1}(-4)$

$\theta = \csc^{-1}(-4)$

$\csc\theta = -4$

$\sin\theta = -\frac{1}{4}$

$\theta = \sin^{-1}\left(-\frac{1}{4}\right)$

$\theta \approx -14.478^\circ$

$\boxed{\csc^{-1}(-4) \approx -14.478^\circ}$

(17) $\sec^{-1}\left(\frac{7}{4}\right)$

$\theta = \sec^{-1}\left(\frac{7}{4}\right)$

$\sec\theta = \frac{7}{4}$

$\cos\theta = \frac{4}{7}$

$\theta = \cos^{-1}\left(\frac{4}{7}\right)$

$\theta \approx 55.15^\circ$

$\boxed{\sec^{-1}\left(\frac{7}{4}\right) \approx 55.15^\circ}$

(18) $\cot^{-1}(5)$

$\theta = \cot^{-1}(5)$

$\cot\theta = 5$

$\tan\theta = \frac{1}{5}$

$\theta = \tan^{-1}\left(\frac{1}{5}\right)$

$\theta \approx 11.31^\circ$

$\boxed{\cot^{-1}(5) \approx 11.31^\circ}$

Different Notation for Inverse Trigonometric Functions

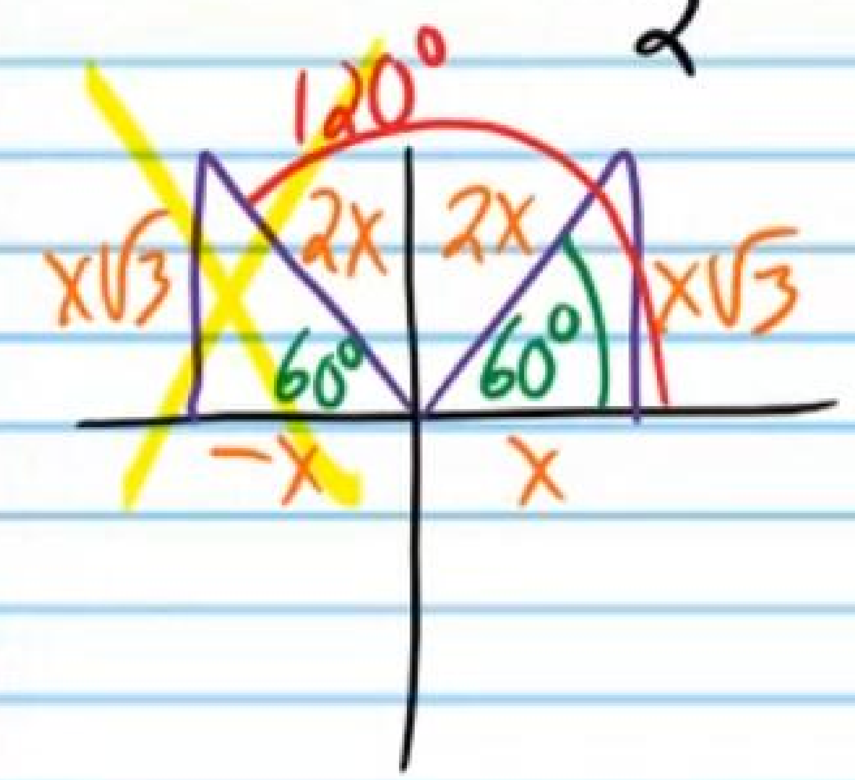
Find the values of the following expressions without a calculator.

(19) $\arcsin\left(\frac{\sqrt{3}}{2}\right)$

$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

$\sin\theta = \frac{\sqrt{3}}{2}$



$\theta = 60^\circ$

$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \boxed{60^\circ}$

(20) $\operatorname{arcsec}(-1)$

$\operatorname{arcsec}(-1) = \sec^{-1}(-1)$

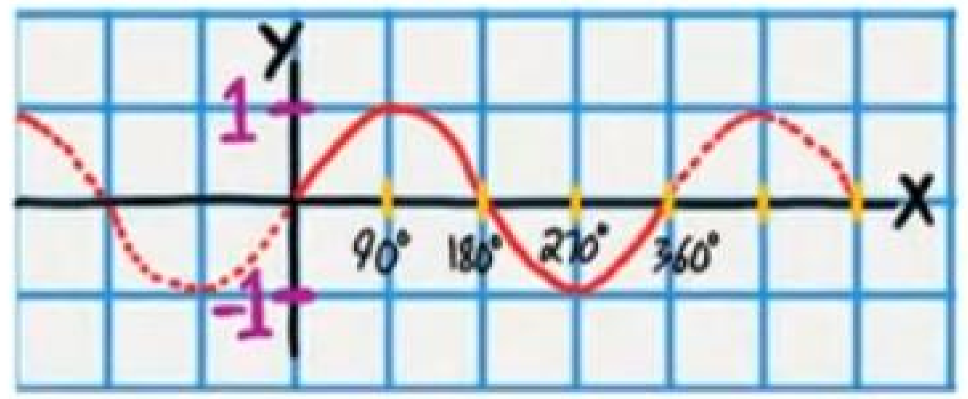
$\theta = \sec^{-1}(-1)$

$\sec\theta = -1$

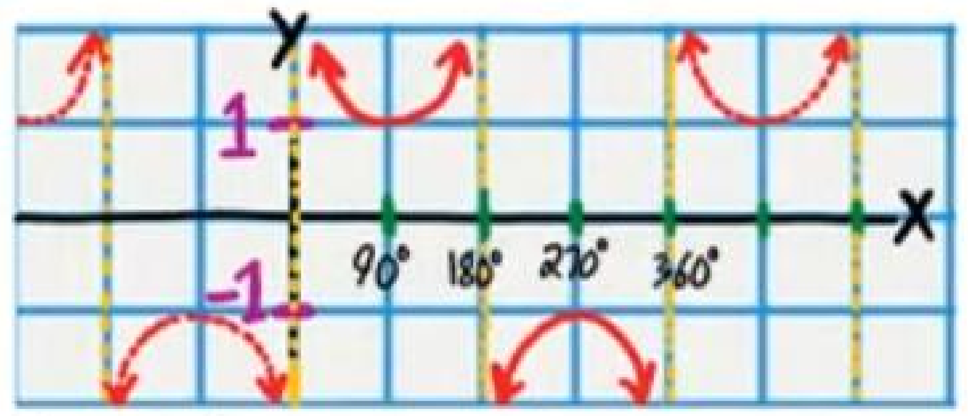
$\theta = 180^\circ$

$\operatorname{arcsec}(-1) = \sec^{-1}(-1) = \boxed{180^\circ}$

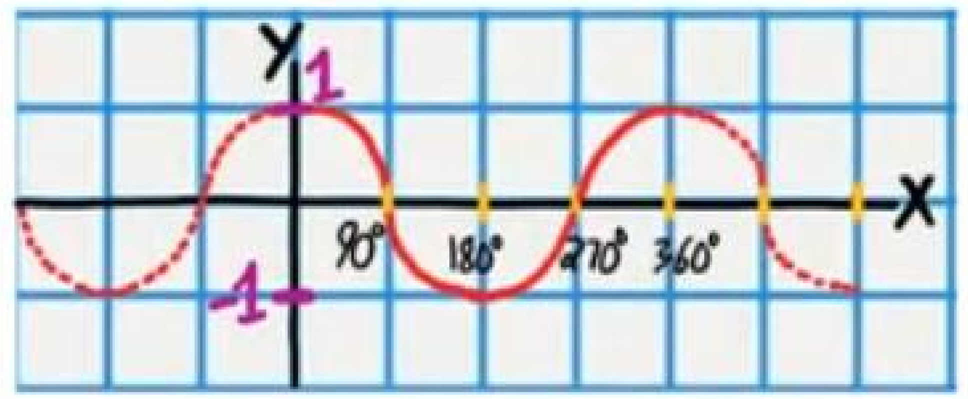
y = sinx



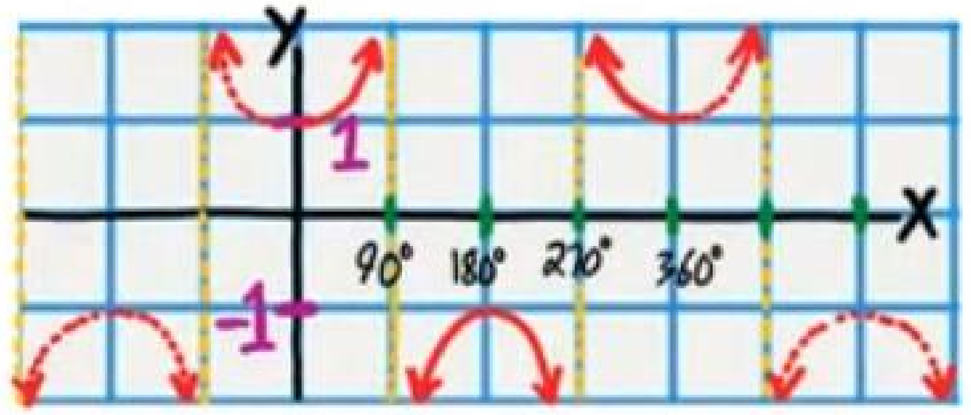
y = cscx



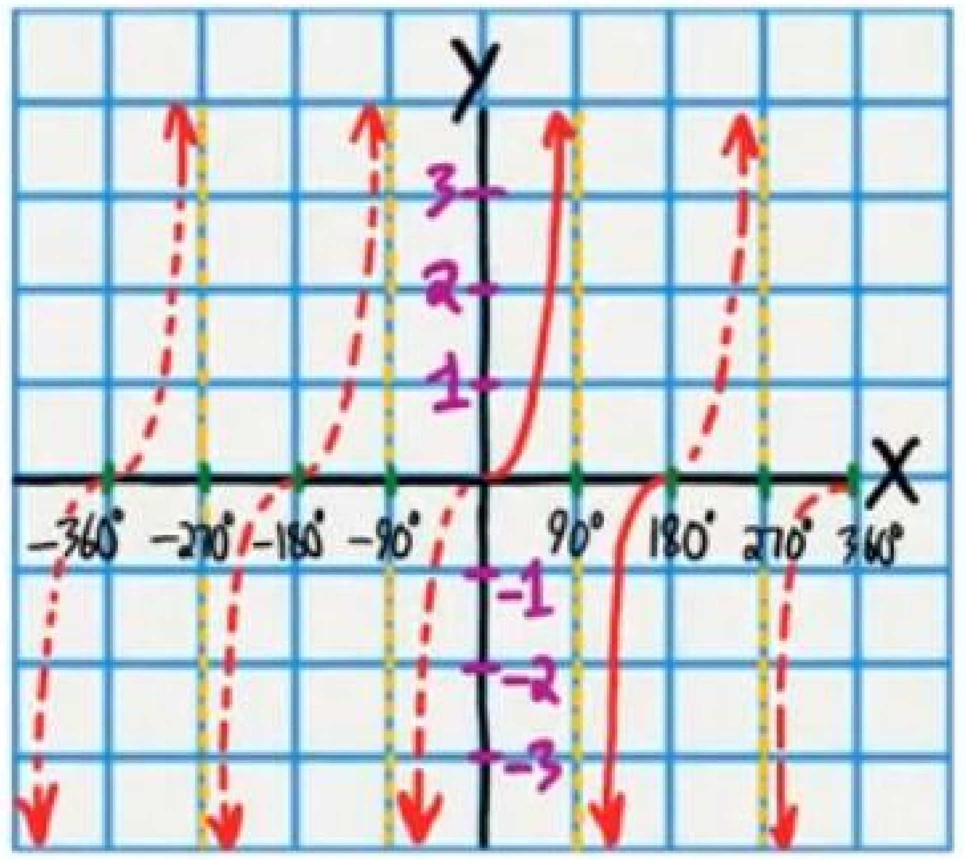
y = cosx



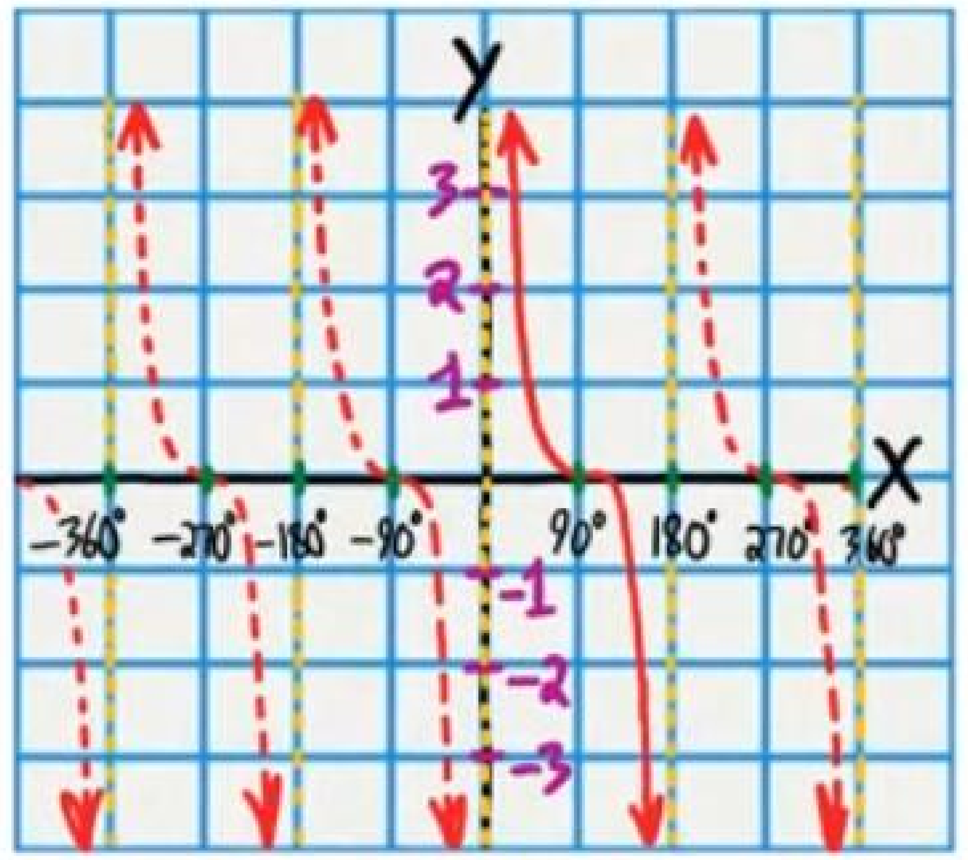
y = secx



y = tanx



y = cotx

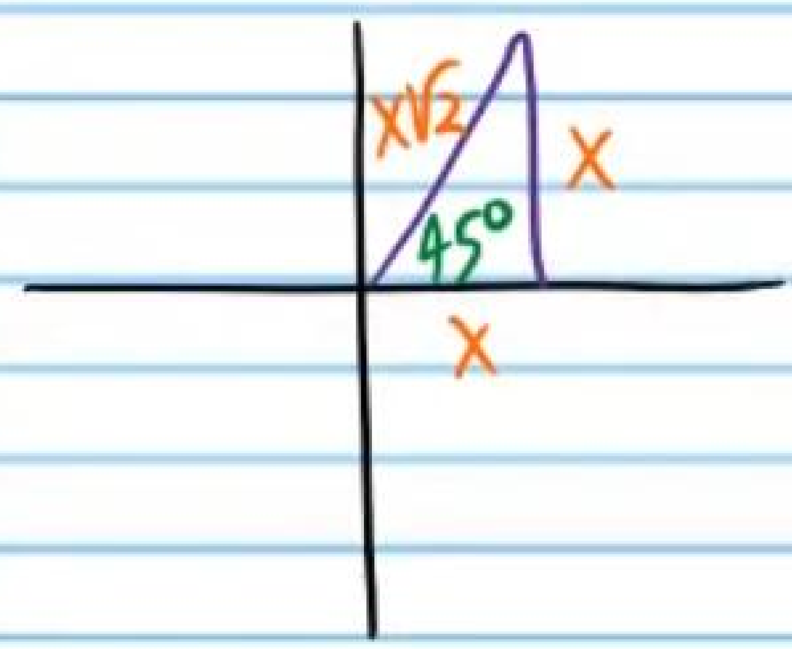


(21) arctan (1)

$\arctan(1) = \tan^{-1}(1)$

$\theta = \tan^{-1}(1)$

$\tan\theta = 1$



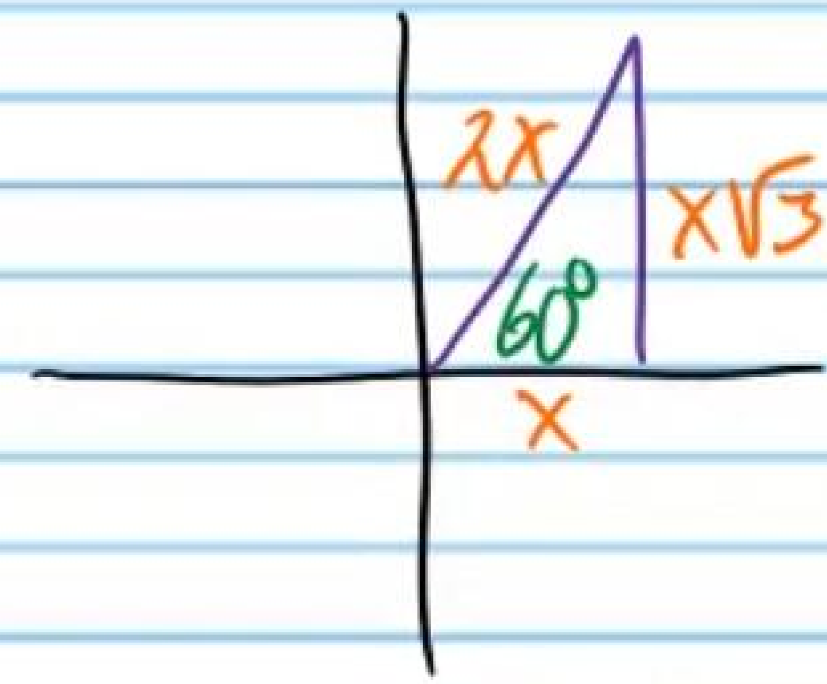
$\theta = 45°$

$\arctan(1) = \tan^{-1}(1) = \boxed{45°}$

(22) arccos $\left(\frac{1}{2}\right)$

$\theta = \arccos\left(\frac{1}{2}\right) = \cos^{-1}\left(\frac{1}{2}\right)$

$\cos\theta = \frac{1}{2}$



$\theta = 60°$

$\arccos\left(\frac{1}{2}\right) = \boxed{60°}$

(23) $\operatorname{arccot}(0)$

$\theta = \operatorname{arccot}(0)$

$\theta = \cot^{-1}(0)$

$\cot\theta = 0$

$\theta = 90^\circ$

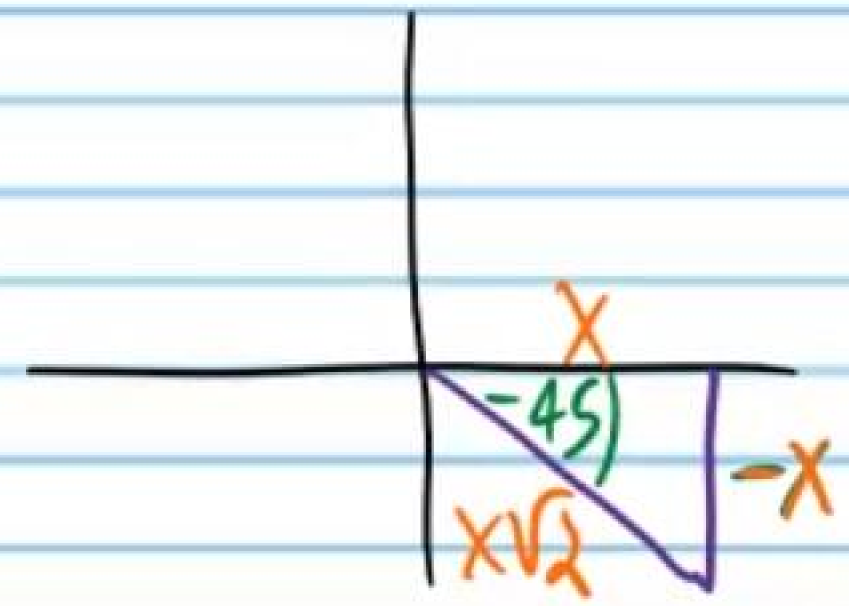
$\operatorname{arccot}(0) = \boxed{90^\circ}$

(24) $\operatorname{arccsc}(-\sqrt{2})$

$\theta = \operatorname{arccsc}(-\sqrt{2})$

$\theta = \csc^{-1}(-\sqrt{2})$

$\csc\theta = -\sqrt{2}$



$\theta = -45^\circ$

$\operatorname{arccsc}(-\sqrt{2}) = \boxed{-45^\circ}$

Solving Trigonometric Equations without a Calculator When Ratios Are Not Recognizable

Do not use a calculator to complete the following problems.

(25) Solve if $0 < \theta < 360^\circ$.

$\sin\theta = -\frac{\sqrt{2}}{2}$

$\sin\theta = -\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{\sqrt{2}}$

$\sqrt{a}\sqrt{b} = \sqrt{ab}$

$\sin\theta = -\frac{\sqrt{4}}{2\sqrt{2}}$

$\sin\theta = -\frac{2}{2\sqrt{2}}$

$\sin\theta = -\frac{1}{\sqrt{2}}$

$\boxed{\theta = 225^\circ, 315^\circ}$

(26) Solve if $0 < \alpha < 2\pi$.

$$\tan\alpha = -\frac{\sqrt{3}}{3}$$

$$\tan\alpha = -\frac{\sqrt{3}\cdot\sqrt{3}}{3\cdot\sqrt{3}}$$

$$\tan\alpha = -\frac{\sqrt{9}}{3\sqrt{3}}$$

$$\tan\alpha = -\frac{3}{3\sqrt{3}}$$

$$\tan\alpha = -\frac{1}{\sqrt{3}}$$

$$\alpha = 150^\circ, 330^\circ$$

$$\alpha = 150^\circ \times \frac{\pi}{180^\circ} = \boxed{\frac{5\pi}{6}}$$

$$\alpha = 330^\circ \times \frac{\pi}{180^\circ} = \boxed{\frac{11\pi}{6}}$$

(27) Solve if $0 \leq W \leq 360^\circ$.

$$\csc W = \frac{2\sqrt{3}}{3}$$

$$\csc W = \frac{2\sqrt{3}\cdot\sqrt{3}}{3\cdot\sqrt{3}}$$

$$\csc W = \frac{2\sqrt{9}}{3\sqrt{3}}$$

$$\csc W = \frac{2\cdot\cancel{3}}{\cancel{3}\sqrt{3}}$$

$$\csc W = \frac{2}{\sqrt{3}}$$

$$\boxed{W = 60^\circ, 120^\circ}$$

(28) Solve if $0 \le A < 2\pi$.

$$-\frac{2}{\sqrt{2}} = \sec A$$

$$-\frac{2 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \sec A$$

$$-\frac{2\sqrt{2}}{\sqrt{4}} = \sec A$$

$$-\frac{2\sqrt{2}}{2} = \sec A$$

$$-\sqrt{2} = \sec A$$

$$A = 135°, 225°$$

$$A = 135° \times \frac{\pi}{180°} = \boxed{\frac{3\pi}{4}}$$

$$A = 225° \times \frac{\pi}{180°} = \boxed{\frac{5\pi}{4}}$$

(29) Solve if $0 < B \le 360°$.

$$-\frac{\sqrt{3}}{3} = \cot B$$

$$-\frac{\sqrt{3} \cdot \sqrt{3}}{3 \cdot \sqrt{3}} = \cot B$$

$$-\frac{3}{3\sqrt{3}} = \cot B$$

$$-\frac{1}{\sqrt{3}} = \cot B$$

$$\boxed{B = 150°, 330°}$$

(30) Solve if $0 \leq \beta \leq 2\pi$.

$$\cos\beta = \frac{\sqrt{2}}{2}$$

$$\cos\beta = \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{\sqrt{2}}$$

$$\cos\beta = \frac{2}{2\sqrt{2}}$$

$$\cos\beta = \frac{1}{\sqrt{2}}$$

$$\beta = 45°, 315°$$

$$\beta = 45° \times \frac{\pi}{180°} = \boxed{\frac{\pi}{4}}$$

$$\beta = 315° \times \frac{\pi}{180°} = \boxed{\frac{7\pi}{4}}$$